A Shot-Noise Limited Fiber Laser Source by Cascaded Passive Optical Filtering

Thanh T-H. Nguyen, Jong H. Chow, Conor M. Mow-Lowry, Malcolm B. Gray, and David E. McClelland

***Abstract*—We present a technique for suppressing relative intensity noise in a fiber laser, which uses two Pound–Drever–Hall feedback control loops to lock two optical fiber ring resonators in series. The resonators act as cascaded passive optical filters to remove high frequency intensity noise. We describe and discuss the control scheme for holding the cavities on resonance, and present the experimental result, demonstrating shot noise limited performance for intensity fluctuations above 2.5 MHz.**

***Index Terms*—Fiber resonators, fiber ring cavity, laser intensity noise suppression, optical filter, Pound–Drever–Hall locking.**

I. Introduction

In applications such as sensing, metrology, and quantum cryptography, where high resolution optical interferometry is involved, a high performance single frequency laser is required [1]–[6]. The performance of a single frequency laser can be measured by its frequency noise and intensity noise. In high resolution interferometry, it is these noise sources which limit the measurement sensitivity. Traditionally, these applications use solid-state lasers, such as the Nd:YAG laser at 1064 nm, as optical sources due to their well-established stability and performance [7]–[9]. However, with the emergence of optical components which benefit from economies of scale at 1550 nm, as well as the maturity of erbium-doped fiber lasers, the second telecommunications window is becoming an important alternative for high sensitivity applications.

Due to its long cavity length and narrow linewidth, a free-running commercial erbium-doped fiber laser can have frequency noise similar to that of a Nd:YAG laser. However, its relative intensity noise (RIN) can extend up to several tens of MHz before shot noise is reached [10]. Reduction of intensity noise at the frequency of interest is important for optical measurements, as it frequently couples into the signal readout to limit sensitivity.

For numerous high end metrology measurements, reducing laser intensity fluctuations to reach the fundamental shot noise limit is the ultimate goal. Shot noise is due to the quantum nature of light and is associated with the random arrival time of photons. It is thus observed as white noise in the frequency domain, with a spectral density given by

$$SN = \sqrt{\frac{2e}{\rho P_{\mathrm{opt}}}} \tag{1}$$

where e is the electronic charge, P_{opt} is the optical power detected, and ρ is the photodetector responsivity (in Amperes per Watt).

In high precision interferometric techniques involving radio frequency laser modulation, such as those used in gravitational wave instrument control [11]–[14] and frequency modulation spectroscopy [15]–[17], the laser source must be limited by shot noise at the modulation frequency for the performance to reach the shot noise limit. On the other hand, in quantum optics experiments such as cryptography, the laser must be shot noise limited across the entire signal band to ensure perfect security.

The most common method to suppress laser intensity noise, particularly that caused by the relaxation oscillation of the laser, is with an active control loop. This technique samples the laser output and provides feedback to the optical pump-diode current, thereby reducing the laser intensity fluctuations. Using active intensity noise suppression, a relative power noise of $2.4 \times 10^{-9}/\sqrt{\mathrm{Hz}}$ at low frequencies (10 Hz) for a cw Nd:YAG laser has been reported [18]. At high frequencies, due to the limited bandwidth of the feedback loop, it is typically effective only up to a few MHz [19], [20]. Any intensity fluctuations above the control loop bandwidth remain unsuppressed. Hence, to generate laser light that is shot noise limited above a few MHz, additional measures must be taken to suppress high frequency intensity noise.

In this paper, we present a method to passively suppress the relative intensity noise of a fiber laser by use of cascaded optical filters. It is effective in reducing relative intensity noise by several orders of magnitude by itself, or can be used in tandem with active intensity stabilization for even better performance. We demonstrated this technique experimentally in an erbium-doped fiber laser source at 1550 nm, and succeeded in suppressing its intensity fluctuations above 2.5 MHz to the shot noise limit. This was achieved by frequency locking two fiber ring cavities in series to the fiber laser.

II. Control of Cascaded Optical Cavities

Fig. 1 shows our conceptual setup where the two cavities are used as noise filters. The laser is kept resonant with the two cavities by two simultaneously active feedback loops. In the first locking loop, cavity 1 is locked to the laser via a piezoelectric-transducer (PZT), which changes the optical path length of

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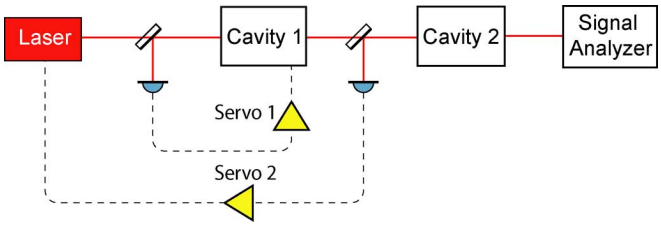

Fig. 1. Conceptual block diagram of the experiment. Servo 1 controls the length of cavity 1, and servo 2 controls the laser frequency.

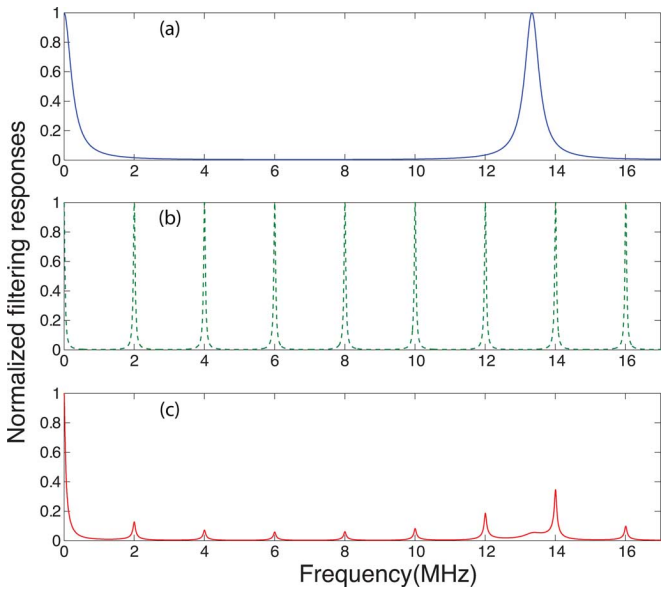

Fig. 2. Theoretical plot demonstrating the idea of the cascaded filtering system. Plots (a) and (b) are the filtering responses of the short and long cavities respectively. Plot (c) is the result of the cascaded system.

cavity 1 to hold it on resonance as the laser frequency changes. The transmitted light from cavity 1 is incident on the second cavity for the next stage of filtering. The correction signal for the second cavity is injected into the laser PZT for frequency tuning in order to keep the laser beam resonant with cavity 2. The output of this cascaded filtering system is the desired RIN suppressed laser light.

The filtering effect of the system is illustrated by the cartoon in Fig. 2. Fig. 2(a) and (b) describe the transmission responses of two cavities of different lengths. Ideally, the resonance peaks of the two cavities should only align at the laser frequency. The resultant filtering function is the multiplication of the two transmission responses in the frequency domain. The solid red line in Fig. 2(c) is the total frequency response of the cascaded filter system.

In the experiment, we designed a dual loop control scheme using the Pound–Drever–Hall (PDH) locking technique [21], [22]. The method requires two sets of radio-frequency (RF) sidebands, one for each cavity. Fig. 3 is a cartoon illustrating the desired relationship between the laser carrier, modulation sidebands, and the two cavity transmission responses. The solid (blue) and dashed (green) peaks are the resonant responses of cavity 1 and cavity 2 respectively. The peaks are misaligned in frequency except for the middle peaks which are aligned with the laser carrier. The carrier and the two sets of sidebands are

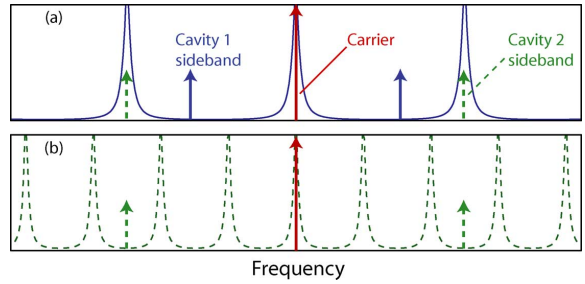

Fig. 3. Illustration describing the dual locking control scheme. The red arrow is the carrier, the solid (blue) arrows are the sidebands for the first cavity, and the dashed (green) arrows are the sidebands for the second cavity. Plot (a) is the transmission response of cavity 1. Plot (b) is the transmission response of cavity 2.

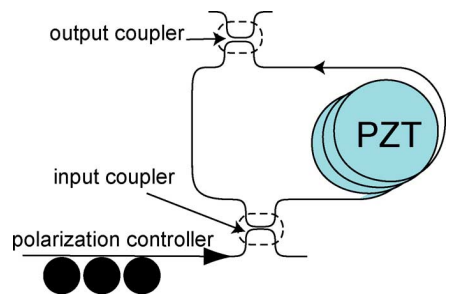

Fig. 4. Illustration of the construction of a typical fiber ring cavity. The inclusion of a PZT is optional and depends on the need for optical path length control in a specific experimental setup.

represented by arrows with the center red arrow signifying the carrier. The two short solid (blue) arrows closest to the carrier, shown in Fig. 3(a), are the sidebands for locking cavity 1 to the laser. These sidebands should ideally be completely reflected off cavity 1, thus their representing arrows are placed at antiresonance with the cavity. The final set of sidebands, shown in both Fig. 3(a) and (b) in dashed (green) arrows, are used in the locking loop of cavity 2. They need to be transmitted by cavity 1, but reflected by cavity 2. Hence, they are resonant with cavity 1, as shown in in Fig. 3(a), but antiresonant with cavity 2, as shown in Fig. 3(b).

III. Experimental Configuration

Fig. 4 illustrates the basic layout of a typical fiber ring cavity. It consists of fusion-tapered couplers at the input and output of the cavity. A length of fiber is fusion spliced between an output port of the input coupler, and an input port of the output coupler. An output port of the output coupler is then spliced to an input port of the input coupler to close the cavity. The output coupler facilitates access to transmitted light.

To maximize cavity finesse, the output port with the smaller coupling ratio is used for coupling into the resonator. Similarly for the second coupler, the smaller coupling port is used for output coupling to optimize circulating power. For example, with a 95/5% input coupler, the 5% port is used for injecting light into the cavity. For the output coupler, the 5% port would be used to tap-off circulating light for transmission. To enable cavity length tuning, part of the intra-cavity fiber can be wound onto a PZT cylinder, whose diameter can be changed by an

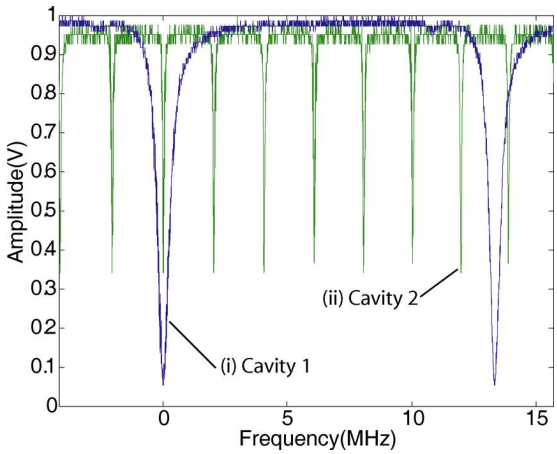

Fig. 5. Reflection responses of the two cavities as a function of frequency, when aligned at the locking point. The green trace is cavity 1 and the blue trace is cavity 2. The cavities were constructed such that the higher order resonant peaks were antialigned.

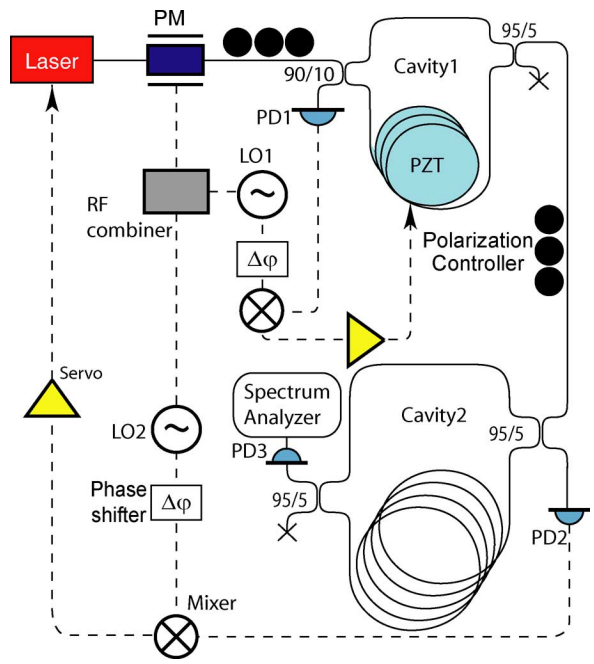

Fig. 6. Detailed graph of the experimental configuration. LO=local oscillation; PM = phase modulator; PD = photodetector.

applied voltage. This stretches the fiber, thereby changing the cavity length, and hence, resonance frequency.

In our experiment, cavity 1 was 15 m long which corresponds to a FSR of about 13.5 MHz. The reflection response of cavity 1 is the blue trace (i) in Fig. 5. It had a full width at half maximum height (FWHM) of 490 kHz and a finesse of 27. A PZT, which had a mechanical resonant frequency at about 22 kHz, was included in this cavity for length tuning. The input and output couplers were 90/10% and 95/5% respectively. The input coupling impedance was matched well with the intra-cavity losses, resulting in minimum reflected power on resonance, as shown in Trace (i) of Fig. 5.

In contrast, cavity 2 was longer in length and relatively undercoupled, resulting in a larger fraction of light reflected on resonance. The reflection response of the second filter is the green trace (ii) of Fig. 5. It was 100 m long with a FSR of 2.05 MHz and FWHM of 60 kHz. This gave the cavity a finesse of $\sim$33. The reflection traces of Fig. 5 were obtained by observing the output of a photodetector while sweeping the laser frequency.

We note that although the finesse of this cavity 2 was low, its linewidth emulates a 20 cm long free-space cavity with a finesse of 12500.

The experimental configuration of the cascaded system is shown in Fig. 6. The two modulation frequencies were generated with two Agilent (Model 33250A) function generators and added using an RF combiner connected to a phase modulator (PM). The locking loop for cavity 1 had a modulation frequency of 50 MHz, which resided near antiresonance. The polarization controller was used to match the input laser beam to an eigenmode of the cavity. The reflected light from cavity 1 was received by photodetector PD1 and converted to an RF electronic signal. It was then mixed with the local oscillator (LO1) to produce an error signal that was optimized by shifting the phase of the local oscillator. This signal, which was proportional to the frequency difference between the laser carrier and the cavity resonance, was used by a servo to generate a feedback voltage. The feedback voltage was applied to the PZT of cavity 1, actuating its optical length to lock its resonance to the laser carrier.

For the second locking loop, the modulation frequency, generated by LO2, was 27 MHz, which was resonant with cavity 1 but antiresonant with cavity 2. As with the locking loop for cavity 1, the RF signal of the second locking loop was received at a photodetector (PD2), mixed down to produce an error signal, and sent to a second servo to produce a feedback signal. This feedback signal was injected into the laser PZT to lock the laser carrier frequency to cavity 2.

For full operation of the system, the two control loops were closed in sequence; first locking cavity 1 then cavity 2. The servo for cavity 1 had a unity gain of approximately 2 kHz with 45° phase margin, providing a stable feedback loop. While the first cavity was locked to the laser, a slow ramp was injected to the laser PZT to scan its frequency relative to cavity 2. This ramp rate had to be within the servo bandwidth of cavity 1 and the ramp amplitude must be less than the dynamic range of the cavity 1 PZT. Once the laser frequency was close to the resonance of cavity 2, the second servo was engaged. This servo had a unity gain of 1.5 kHz and a phase margin of 30°. The gain for this feedback loop was increased at low frequencies in order to hold the laser on resonance more effectively. Both cavities were placed inside anechoic chambers to mitigate environmental perturbations, such as mechanical noise and temperature fluctuations.

IV. Results

Fig. 7 displays the relative intensity noise at the output of the cascaded filters operating with a detected power of 350 μW. The unfiltered RIN of the free-running laser is shown as trace (i), with the relaxation oscillation at 1.2 MHz. It was obtained directly from the laser source prior to the filtering cavities with an Agilent *E*4402*B* RF spectrum analyzer. The electronic noise,

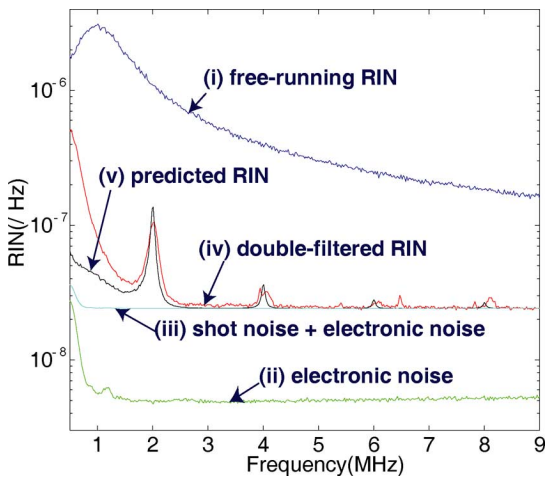

Fig. 7. The experimental result of the cascaded filtering system. The relative intensity noise is shot noise limited above 2.5 MHz, with small residual peaks due to the filtering limitation of cavity 1.

trace (ii), was the output of the photodetector response in the absence of incident light. Trace (iii) shows the quadrature sum of electronic noise and calculated shot noise. For 350 μW of optical power and assuming photodetector responsivity of 1 A/W, the shot noise calculated using (1) is about $3 \times 10^{-8}/\sqrt{\text{Hz}}$. The experimental result after the two passive filters is displayed as trace 7 (iv). The filtered RIN after the cascaded ring cavities approaches the shot noise limit above 2.5 MHz. Below this frequency, the intensity noise is limited by the filtering response of the cascaded cavities. The largest noise reduction obtained in this setup is 34 dB, near the relaxation oscillation frequency of 1.2 MHz. At 2 MHz and 4 MHz, we note two small peaks. There, the laser RIN is suppressed by 17 dB, limited by the filtering response of the short cavity. Trace (v) is the predicted filtered RIN, which was obtained by multiplying the free-running RIN, shown as trace (i), with the theoretical filtering response of the cascaded system, represented in plot (c) of Fig. 2. The experimental result is in good agreement with the theoretical predictions at frequencies above 2 MHz. At low frequency, the measured RIN appears to be higher than predicted. This is likely a result of frequency noise coupling into intensity noise at low signal frequencies.

V. Limitations and Design Considerations

The main advantage of using fiber ring resonators as passive filters is the ability to attain very narrow resonances with relatively low cost components by using long lengths of intra-cavity fiber. We had noted in Section III that the fiber cavities described in this experiment emulated the resonance full-width half-maximum of a free space resonator with extremely high finesse, without the cost of high quality optics and vacuum housing required. However, fiber ring cavities pose some limitations on our system performance.

In order to control the two fiber ring cavities, the laser beam was phase modulated at two frequencies with a modulation depth of 0.04 radians. These sidebands were suppressed after transmission through the two cavities. With the control scheme used in this experiment, the first set of sidebands (at 50 MHz) was near the antiresonance condition of both cavities. It therefore experienced maximum, or close to maximum, suppression. The residual power in each sideband of the 50 MHz sidebands was approximately 7×10^{-12} W, which is equivalent to a modulation depth of 2×10^{-6} radians. The second set of sidebands (at 27 MHz) were near the maximum transmission frequency of the first cavity and nearly antiresonant with the second cavity. The residual power in these sidebands was 1.3×10^{-10} W or 3.5×10^{-5} radians.

For most fiber sensing [2], metrology, and quantum optics systems [23], [24], an optical power of around 1 mW is ample. There are, however, applications where higher power is desirable. The primary limit to our laser output power is the stimulated Brillouin scattering (SBS) threshold in the long fiber length of cavity 2. The effects of SBS are exacerbated due to its resonant circulating power.

The easiest way to facilitate higher optical output power is by reducing the length of cavity 2 to increase its SBS threshold. However, for a given cavity finesse, this would increase the resonance FWHM of cavity 2 and degrade its effectiveness at filtering low frequency noise, thus 1) reducing the suppression of the relaxation oscillation; and 2) increasing the frequency corner where the laser reaches shot noise. This degradation in roll off can partly be offset by simultaneously increasing the length of cavity 1, as long as reasonable modulation frequencies can be found to control the cavity lengths. One possible alternative is to use the new generation of large core fibers instead of SMF-28, which reduces the laser intensity in the core, thus increasing the SBS threshold.

The technique described in this paper is highly effective in reducing relative intensity noise in a laser, regardless of whether reaching shot noise performance is the aim. At present, the suppression function, or the ratio between maximum (resonant) and minimum (anti-resonant) transmission, of the cascaded filters is sufficient to reduce the intensity noise to the shot noise level for our laser output power. If output power is increased, the relative shot noise will decrease according to (1). In that case, the laser RIN will no longer be limited by shot noise, but rather by the suppression function of the filters. If shot noise performance is the primary goal, at higher optical powers the finesse, especially of cavity 1, must be improved to increase the suppression ratio. The finesse of our fiber ring cavities are primarily limited by losses in the fusion-tapered fiber couplers. These losses can be reduced by using high quality evanescent couplers. We note that increasing cavity finesse, increases circulating power. Care must therefore be taken to stay within the SBS threshold of cavity 1 at higher finesse.

VI. Conclusion

We have presented a technique to passively suppress the relative intensity noise of a laser by using a system of cascaded fiber resonators as optical filters. The cavities were simultaneously held on resonance using two Pound–Drever–Hall feedback control loops. We discussed the design of the optical filters, as well as the system control scheme. By implementing this technique, we have produced an erbium doped fiber laser source with shot noise limited relative intensity noise above 2.5 MHz.

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